MicroPulse Coach Guide Series

Sprint Exposure (IMA)

Weekly sprint load · Hamstring injury risk · Periodization

May 2026 · v1.0

Quick summary

Sprint Exposure measures **how much high-velocity sprint work a player did in training over the last 7 days, expressed as a ratio of what they typically do in a single match**. Too little (< 50%) raises hamstring injury risk ~3× (Malone 2018). Too much (> 180%) is an accumulated spike that warrants reduced load over the next 1–2 days.

The metric uses **IMA Free Running** (Catapult Vector S7+ pod), which detects every footstrike via the on-board IMU and buckets each stride into one of 8 velocity bands. MicroPulse sums **bands 5–8** (every stride at 18+ km/h) and calls that the **sprint-stride count** for that day.

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1. What you see in the system

The chip in Decision Summary (Daily Briefing)

Sprint Exposure appears as an **informational chip** below each player's row in Decision Summary. The chip is **not verdict-changing** — it doesn't push a player into RECOVERY by itself. The coach decides whether to act.

Example chip text:

- **Not enough sprinting this week (only 35% of match demand)** — Add a sprint block at the next session.
- **Sprint volume slightly below normal (72% of match demand)** — Top up a bit if a match is coming up.
- **Too much sprinting this week (212% of match demand)** — Cap sprint volume the next 2 days.

The widget on /coach/decel-intelligence

A larger view of the same metric with:

- **Visual progress bar** from 0% to 150% with reference lines at 50/80/100%.
- **Week (7d)** — total sprint-strides accumulated in training over the last 7 days.
- **Match demand** — the player's personal baseline (average sprint-strides per match).

- **Days observed** — how many days of the last 7 had data.

2. Sport-science background

Key study: Malone et al. 2018

Malone and colleagues tracked 14 professional teams across two seasons and found a clear **U-shaped relationship** between weekly high-speed running exposure and hamstring injuries:

- **< 50% of match demand**: 3× higher hamstring injury risk. Undertraining leaves the hamstring unaccustomed to eccentric loading at sprint speeds.
- **80–130% of match demand**: lowest injury risk — the sweet spot.
- **> 150% of match demand**: rising risk again, due to accumulated tissue stress.

Both ends of the U-curve are dangerous. In practice, most club teams undertrain rather than overdose sprints, especially during congested fixture periods.

Supporting research

Buchheit & Cardinale 2019 (Sport Performance & Science Reports): Recommend monitoring *both* QUALITY (peak sprint velocity drop — see Sprint Speed Drop chip) and VOLUME (sprint exposure) together, not either alone. Quality captures mechanical fatigue on a single day; volume captures chronic under/overload across the week.

Edouard et al. 2019: A Sprint Speed Drop > 10% below personal max is an early warning sign for hamstring strain within the next 24 hours. Sprint Speed Drop is the QUALITY companion to Sprint Exposure (which is the VOLUME side).

McBurnie et al. 2022: Demonstrate that IMA-based stride-count measurements capture **eccentric mechanical load** that GPS velocity-band metrics miss — especially during change-of-direction (COD) and agility work.

Bishop et al. 2020: L/R asymmetry > 15% is the strongest single predictor of non-contact lower-limb injury. This is shown as a separate chip in MicroPulse and complements Sprint Exposure.

3. Why IMA instead of GPS velocity bands?

This is the most important conceptual point in this guide. MicroPulse uses IMA stride-count as the foundation for Sprint Exposure, not GPS distance in V5/V6 bands. The two methods capture genuinely different things.

IMA Free Running (Inertial Movement Analysis)

- **Where:** Inside the Vector S7+ pod (gyroscope + accelerometer).
- **Sampling rate:** 100 Hz (100 measurements per second).
- **What it measures:** Every footstrike, the velocity at each stride, and buckets each stride into one of 8 velocity bands.
- **Works indoors:** Yes — no GPS signal needed.

- **Captures:** Stride count at high cadence/velocity, including quick-cadence work during COD and agility.

GPS Velocity Bands (V5, V6)

- **Where:** From the GPS receiver in the pod.
- **Sampling rate:** 10 Hz (10 measurements per second).
- **What it measures:** Distance covered in each velocity band: V5 = 19.8–25 km/h, V6 = > 25 km/h.
- **Works indoors:** No — no GPS signal = no data.
- **Captures:** Forward-velocity distance and the count of sprint bursts (efforts).

Why the difference matters — a real example

Let's look at Óli Valur Ómarsson on Monday 11 May (training day):

Metric	Value	Interpretation
Total Distance	4,959 m	Full training session
Player Load	451	Medium-to-high accelerometer load
GPS V5 distance	105 m	Very little high-speed running
GPS V6 distance	14 m	Almost no max-velocity sprinting
GPS V5+V6 efforts	13	13 separate sprint bursts
IMA bands 5–8 strides	**629**	Large volume of high-cadence strides

How can a player register **629 sprint-strides** when GPS shows only 119 m of sprint distance and 13 sprint bursts?

The explanation: IMA Free Running uses an IMU-derived velocity estimate (stride rate × stride length) — not GPS. On a possession-heavy training day with rondos, 1v1s, and small-sided games, Óli Valur took a **huge number of short, quick steps** (quick shuffles, acceleration steps, side-stepping). The IMU detects each footstrike and rates the stride rate as high; the algorithm buckets these into bands 5–8 even though the player isn't actually travelling forward at 18+ km/h.

Doing the math:

- 13 sprint efforts × ~7 strides per effort ≈ **~91 "true" sprint-strides** (matches GPS V5+V6).
- 629 − 91 ≈ **~538 IMA-strides from quick-cadence / agility / COD work** that GPS doesn't see.

Why this is a strength, not a weakness

Hamstring injuries don't only happen during long sprints. McBurnie 2022 and Edouard 2019 show that **a large fraction of hamstring strains occur during quick-direction work, accelerations, and terminal-swing steps** — all of which GPS misses because the events are too brief.

IMA captures **eccentric mechanical load** regardless of whether the player is running in a straight line. That makes the MicroPulse Sprint Exposure metric closer to "high-velocity mechanical load exposure" — a richer, more injury-relevant signal than pure sprint distance.

4. How the system calculates Sprint Exposure

The calculation is fully deterministic (no AI) and grounded in sport-science research. Here it is step-by-step:

Step 1: Acute load (last 7 days, training-only)

MicroPulse sums **IMA bands 5–8 strides** across all **training days** in the last 7 days. Match days are excluded from this sum (see Step 2 for why).

Why training-only? The match itself is already in the baseline (denominator). Including it in the acute sum (numerator) would double-count the match, artificially pushing every active player above 100% and triggering false "Spike" verdicts. This was a real bug we found in our first version and fixed in May 2026 after seeing every Breiðablik player flagged as Spike.

Step 2: Chronic baseline (match demand, last 28 days)

MicroPulse computes the **average IMA bands 5–8 strides per match** across all matches the player played in the last 28 days.

Match days are detected from two sources, in order of priority (UNION):

- **Week plans:** The team schedule (`day_type = 'GAME'`). This is authoritative — the system trusts the weekly plan.
- **Match minutes:** Players who logged ≥ 30 min on that day in per-player minutes records.

Step 3: GPS V5+V6 efforts fallback (when IMA data is missing)

Catapult fixed an IMA Free Running config issue in May 2026 — but older activities can't be re-processed. On those historical match days IMA bands 5–8 are all 0, even though GPS V5+V6 efforts are present.

The algorithm:

- 1. Finds one IMA-complete match day for each player (where both IMA and GPS efforts exist).
- 2. Computes the player's personal **strides-per-effort ratio** = IMA b5-8 ÷ V5+V6 efforts. This is individualised — it reflects each player's cadence profile.
- 3. On other match days where IMA is missing but GPS efforts exist: **estimated IMA strides = V5+V6 efforts × the player's personal ratio**.

Why match-only calibration? On training days, cadence-to-velocity ratios vary wildly (lots of agility = very high strides-per-effort). On matches, ratios are stable across players because the game itself is the constant — players sprint similar distances at similar paces. Calibrating only on match days keeps the estimate accurate.

Example (Höskuldur): 2026-05-08 was an IMA-complete match (IMA 660 / efforts 61 = 10.82 ratio). On the 2026-04-22 match IMA = 0 but GPS V5+V6 efforts = 56. Estimated IMA = 56 × 10.82 = **606 strides**.

Step 4: Compute ratio and assign band

Exposure Ratio = Acute (7d) ÷ Match Day Demand

The band is assigned from Malone-2018-derived thresholds with empirical tuning for football weeks:

Band	Range	Colour	Meaning
UNDERLOAD	< 50%	Red	Undertraining. Hamstring not conditioned to high-velocity load — 3× injury risk (Malone 2018).
WATCH	50–80%	Yellow	Slightly under-loaded — playable but undertrained. If a match is coming, add a sprint block.
SAFE	80–180%	Green	Healthy weekly sprint load. Sweet spot for hamstring protection.
OVERLOAD	> 180%	Red (Spike)	Accumulated spike. Cap sprint and high-cadence work the next 2 days.

A note on the SAFE range. *It runs to 180%, not 130%, even though Malone's original paper suggested 80–130%. The reason is empirical: a typical football week has 3–4 training days plus one match. Each training day delivers roughly 50–60% of one match's worth of sprint-relevant work, so a healthy week sits at 130–160% of one match. Setting the SAFE ceiling at 130% would flag every normal football week as a Spike — which contradicts the data. After tuning against Breiðablik historical weeks, 180% emerged as the threshold where ratios truly become biologically meaningful.*

5. Practical examples — Breiðablik 1st team

Two players from the same squad with nearly identical training weeks but very different verdicts. This is per-player calibration doing its job.

Example 1: Kristófer Ingi Kristinsson — 273% Spike

Day	Type	V5+V6 efforts	IMA b5-8 strides
Mon 11 May	Training	14	**434**
Fri 8 May	**MATCH**	60	**274**
Thu 7 May	Training	10	**343**
Wed 6 May	Training	9	*0 (pre-fix)*

Calculation:

- Acute 7d (training-only): 434 + 343 + 0 = **777 strides**
- Match-day demand baseline: **285 strides** (1 measured + 4 GPS-estimated matches)
- Ratio: 777 ÷ 285 = **273%** → **SPIKE (red)**

Interpretation: Kristófer has a low match-day baseline (only 285 strides per match), which suggests a positional player — probably a centre-back or holding midfielder — who doesn't accumulate large sprint-stride volumes in games. His training week delivered 777 strides, **2.7× his typical match's worth** of high-cadence work. The system is flagging that this week was unusually demanding for his profile.

Action: Reduce sprint and high-cadence work for the next 2 days. He doesn't need to be pulled from training — just shift the focus. Fewer rondos in tight spaces, fewer pure agility drills, more ball-possession work at controlled tempo. Re-evaluate after 2 days.

Example 2: Óli Valur Ómarsson — 172% Safe

Day	Type	V5+V6 efforts	IMA b5-8 strides
Mon 11 May	Training	13	**629**
Fri 8 May	**MATCH**	97	**562**
Thu 7 May	Training	8	**292**
Wed 6 May	Training	9	*0 (pre-fix)*

Calculation:

- Acute 7d (training-only): 629 + 292 + 0 = **921 strides**
- Match-day demand baseline: **534 strides** (1 measured + 4 GPS-estimated matches)
- Ratio: 921 ÷ 534 = **172%** → **SAFE (green)**

Interpretation: Óli Valur's match-day baseline is much higher (534 strides — nearly 2× Kristófer's). That suggests a winger or attacking midfielder with high-cadence demands in games. The same training week delivers him only 1.72× his typical match — a healthy weekly load that matches his game profile.

Action: None. Healthy week for his position and playing style.

What these examples teach

Same training week, very different verdicts. **Per-player calibration is working correctly:**

- The same training session loads different players differently.
- Each player's baseline reflects their position and playing style.
- A ratio above 100% is not automatically a problem — it depends on personal baseline.

Without per-player calibration, a team-wide threshold would mislead on at least one of these two players (either hiding Kristófer's spike or falsely flagging Óli Valur as overloaded).

6. Coach actions by band

UNDERLOAD (< 50%) — Undertrained, elevated injury risk

Priority: Add a sprint block at the next session (MD-3 or MD-4 ideal).

Practical inputs (15–20 min):

- 3 × (3 × 30 m accelerations) with 1 min rest between reps, 3 min between sets.
- Or: 4 × (40 m progressive sprint, 50–100% effort) with full recovery.
- Progressive build-up — don't jump from 0 to 100% in one week. Add one sprint day at a time over 1–2 weeks.

Sport-science rationale: Malone 2018 found 3× higher hamstring injury risk in players below 50% of match demand. The hamstring needs eccentric loading to maintain protective capacity (Bahr 2015 Nordic-curl research).

WATCH (50–80%) — Under-loaded, not critical

Priority: Top up if a match is coming. If no match is near, you can hold steady.

Practical inputs (5–10 min):

- Short sprint block at end of session: 4 × 20 m, 80–90% effort, full recovery.
- Or integrate into small-sided games on a larger pitch with width to force high-velocity running.

SAFE (80–180%) — Sweet spot

Priority: No specific action. Continue the current programme.

Still monitor:

- Sprint Speed Drop (separate metric) — if peak velocity drops > 10%, ease off sprint volume regardless of the SAFE band.
- L/R Asymmetry — if asymmetry > 15%, work on CoD/agility before adding more sprint load.

OVERLOAD (> 180%) — Spike, caution warranted

Priority: Reduce sprint and high-cadence work for the next 2 days.

Practical inputs:

- Skip max-effort sprints for the next 1–2 sessions.
- Reduce possession work in tight spaces (where cadence is highest).
- Shift focus to technical work — passing patterns, set pieces.
- Re-evaluate after 2 days — the chip should move to SAFE or WATCH as the acute sum declines.

Sport-science rationale: Malone 2018 found > 150% is associated with accumulated tissue stress and rising injury risk. Our tuned threshold of 180% reflects realistic football-week patterns while still flagging genuine spikes.

7. Confidence, limitations, and when to trust less

Confidence labels in the chip

The chip and widget surface confidence transparency so you know how much to trust the verdict:

- **"Based on 5 matches (3 measured + 2 GPS-estimated)"** — high confidence. Baseline built on 5 matches, 3 with real IMA data.
- **"Based on 3 matches (1 measured + 2 GPS-estimated)"** — medium confidence. Cross-check with Sprint Speed Drop (quality).
- **"Based on 1 of 5 matches (others missing data)"** — low confidence. Data gaps (DNP / pod failures). Interpret with caution; combine with Sprint Speed Drop and wellness data.
- **"Only 1 match in baseline — interpret with caution"** — new player or Catapult just rolled out. Wait another week before acting on harder verdicts.

When not to trust the metric

- **First 28 days on Catapult:** Baseline still being built. Stick with coach intuition until ≥ 3 matches are in the baseline.
- **Returning from serious injury:** The player's old baseline doesn't reflect their current capacity. Re-baseline over 2–3 matches before trusting SAFE / SPIKE verdicts.
- **Position change:** If a player moves from defence to winger (or vice versa), their old-position baseline is invalid. Re-baseline.
- **Indoor training without GPS:** If GPS signal is poor (indoors or under cover), prefer the Indoor Load page over GPS V5/V6 interpretation.

Cross-checks to use

Sprint Exposure works best when read **together with** other signals:

- **Sprint Speed Drop** (quality side): If peak velocity is > 10% below personal max AND Sprint Exposure is SPIKE → caution is essential.
- **Player Load ACWR** (Gabbett 2017): If ACWR > 1.5 AND Sprint Exposure is SPIKE → double risk factor.
- **L/R CoD Asymmetry** (Bishop 2020): If asymmetry > 15% AND Sprint Exposure is SPIKE → both quality and volume risk.
- **Wellness questionnaire:** If the player reports high fatigue (RPE > 7) AND SPIKE → high probability of injury in the next 24–48 hours.

8. Frequently Asked Questions

Why does the same training session show different risk levels for different players?

Per-player calibration. Each player has their own personal baseline (average sprint-strides per match) that reflects their position, playing style, and physical pattern. The same training session can be 2.7× match-load for a centre-back (low match baseline) and 1.7× match-load for a winger (high match baseline).

Why does the same player keep showing up as Spike week after week?

Three possibilities:

- **Chronically under-loaded in matches:** The player regularly comes on as a sub and plays limited minutes, which gives a low sprint-stride count in matches. Every full-volume training week then looks like a "spike" relative to that low baseline.
- **Per-player calibration is still noisy:** Only 1 measured match in the baseline. Wait for more match data before acting on hard verdicts.
- **Genuine overtraining:** Rarer, but possible. Cross-check with Foster Strain and Player Load ACWR.

Why is there sometimes such a big gap between IMA and GPS numbers?

On training days with quick-cadence work (possession, agility, COD), IMA numbers are often 5–10× higher than GPS numbers. This is normal — IMA captures quick-cadence work that GPS misses. It does NOT mean IMA is misleading; it means IMA is a richer measurement of biomechanical load.

On match days (where movement patterns are more straight-line running) IMA and GPS numbers agree much more closely.

What if the player forgot their pod that day?

That day is skipped in the calculation (no data). If multiple days in a row are skipped:

- Acute (7d) becomes lower — the player could fall into UNDERLOAD / WATCH despite having trained.
- Baseline (28d) updates less often — reduced sensitivity to recent matches.

Workaround: A manual entry with approximate values (TD, PL, V5/V6 distance) works for other metrics but provides no IMA data. The system uses the manual entry in distance-based metrics (Player Load ACWR, Foster Strain) but skips it for Sprint Exposure until pod data returns.

Why is SAFE 80–180% and not 80–130% as Malone 2018 described?

Malone 2018 (and most studies) are based on professional teams with very tightly periodised weeks. Their acute/chronic ratio sits closer to 1.0. In our data (Icelandic football), a typical healthy week has 3–4 training sessions plus one match, which produces an acute/chronic ratio around 1.3–1.6 (130–160%). That's **not a spike** — that's normal.

After empirical tuning against Breiðablik data (May 2026): SAFE max was moved from 1.5 (150%) to 1.8 (180%) to reflect real weekly patterns. The thresholds remain sport-science-grounded but are calibrated for Icelandic football context.

Can this metric be used for sports other than football?

Yes — IMA Free Running works for any sport that involves running (football, rugby, handball, athletics, lacrosse). Thresholds need calibrating per sport — e.g. handball has much more COD work and higher quick-cadence than straight-line sprinting. For indoor sports (basketball, volleyball) we'd need indoor-mode thresholds that account for the different movement profile.

9. References and further reading

Key research

Malone S, Hughes B, Doran DA, et al. (2018). Can the workload–injury relationship be moderated by improved strength, speed and repeated-sprint qualities? *J Sci Med Sport* 21(7):710–715.

Buchheit M, Cardinale M. (2019). Adding heat to the live-load monitoring debate. *Sport Performance & Science Reports* #59.

Edouard P, Mendiguchia J, Lahti J, et al. (2019). Sprint mechanical properties and acute hamstring injury risk in elite football. *Br J Sports Med* 53:1330–1336.

McBurnie AJ, Harper DJ, Jones PA, et al. (2022). Deceleration training in team sports. *Sports Med* 52:1473–1487.

Bishop C, Read P, Bromley T, et al. (2020). The association between inter-limb asymmetry and injury risk in soccer players. *J Athl Train* 55:1062–1070.

Bahr R, Thorborg K, Ekstrand J. (2015). Evidence-based hamstring injury prevention. *Br J Sports Med* 49:1466–1471.

Technical references

Catapult Sports (2024). Vector S7+ IMA Free Running V2 — Algorithm Documentation. Catapult Cloud documentation.

Buchheit M, Allen A, Poon TK, et al. (2014). Integrating different tracking systems in football. *J Sports Sci* 32(20):1844–1857.

Related MicroPulse guides

- MicroPulse Coach Guide: Decel Intelligence (McBurnie 2022)
- MicroPulse Coach Guide: Quadrant View (Gabbett ACWR)
- MicroPulse Coach Guide: Indoor Load (FMP / Brown 2016)
- MicroPulse Catapult-IMA Briefing (technical deep-dive)

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